MODELING NORMALCY: A ZERO-NEGATIVE-SHOT DEFENSE AGAINST PROMPT INJECTION

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ABSTRACT

System prompts are the primary mechanism for customizing and constraining large language models (LLMs) to task-specific constraints, yet they are vulnerable to prompt injection. In practice, developers typically have many benign messages for a fixed system prompt, but few or no labeled attacks. Moreover, violations are prompt-specific, necessitating prompt-specific defenses. To address this, we introduce TRAP-BENCH, a realistic, diverse, and large-scale benchmark that maps granular constraints to representative system prompts, paired with benign interactions and constraint-specific attacks. Building on TRAP-BENCH, we present DIAMOND (*Detecting Injections via Activation MONitoring & anomaly Detection*), a zero-negative-shot detector that frames prompt injection as anomaly detection in activation space, modeling hidden-state regularities during normal use and flagging deviations at inference time. Evaluated on TRAP-BENCH, DIAMOND substantially outperforms a strong baseline on **88–98%** of tasks across multiple models and achieves a task-averaged macro F1 of **0.75–0.83**. This robust performance holds even at low false positive rates, confirming its practical viability. Together, our contributions provide both a realistic testbed for evaluating defenses and a practical, drop-in monitor for deployment.

1 INTRODUCTION

Large language models (LLMs) are increasingly the engine for specialized applications, from customer service bots to code assistants (Minaee et al., 2025). In these systems, the system prompt acts as a foundational contract: a set of developer-defined instructions that constrain the model's behavior to a specific role, format, and set of guardrails (Jeoung et al.; Zou et al., 2024). The fragility of this contract, especially when facing conflicting user inputs or complex instructions, creates a critical vulnerability to prompt injection (Mu et al., 2025). Such attacks, which subvert developer directives, lead to significant downstream harm and represent the top risk in the OWASP Top 10 for LLM Applications (OWASP GenAI Security Project, 2025). While related to general jailbreaking, our focus is on detecting violations of developer-defined operational constraints (the system prompt), rather than bypassing the model's foundational safety training.

While various defenses have been proposed (Lla; Chen et al., 2025), they do not directly address the task-specific security setting. In a live application, developers possess a specific system prompt and a wealth of benign user interactions, but have few, if any, labeled attacks. This operational reality highlights a fundamental mismatch. The challenge is not ensuring generic safety, but enforcing adherence to a specific application contract, where violations are defined by the prompt itself.

This gap between research assumptions and deployment needs is mirrored in existing benchmarks. Previous work has evaluated rule-following in synthetic scenarios (Mu et al., 2024), and general benchmarks for adversarial attacks exist (Zou et al., 2023), but there remains a need for a testbed that combines realistic system prompts with the asymmetric data conditions developers actually face.

To address this, we introduce a framework for task-grounded, unsupervised prompt injection detection. Our central insight is to reframe the problem from supervised classification to anomaly detection. Instead of learning the features of known attacks, our approach learns the manifold of internal LLM activations that characterize compliant behavior, flagging any significant deviation. This paper presents two core contributions built on this paradigm: the TRAP-BENCH benchmark, designed

to model these realistic evaluation conditions, and DIAMOND, a zero–negative–shot detector that makes this approach practical for deployed systems.

Contributions

- We introduce TRAP-BENCH, a benchmark of 50 real-world tasks designed to model realistic deployment conditions, featuring a fixed system prompt, abundant benign data, and attacks annotated against specific constraints.
- We propose DIAMOND, a practical, zero-negative-shot detection method that frames prompt injection as anomaly detection in an LLM's latent space. By training a lightweight autoencoder solely on the hidden states of compliant completions, it identifies attacks as deviations from normal processing without requiring any attack examples.
- We conduct an extensive evaluation across seven models (4B–70B) and all 50 tasks, demonstrating that our latent-space detector consistently outperforms strong baselines, establishing the viability of unsupervised monitoring for defending against prompt injections.

2 RELATED WORK

Our work addresses the subversion of an application's system prompt, a practical challenge where developers have abundant benign interactions but few, if any, labeled attacks. We situate our contributions by reviewing prior work on LLM defenses and the evolution of task-specific evaluation.

LLM Defenses and Their Data Dependencies. Defenses against prompt injection have historically centered on methods requiring curated data. **Input/output filters** (Lla; ProtectAI.com, 2023) and **adversarial training** (Chen et al., 2025) are effective against known attack patterns but are less suited to the "zero-negative-shot" reality of a newly deployed application. More recent work has shown that robustness can be improved through adversarial fine-tuning (Mu et al., 2025; Chen et al., 2025; Chen et al.). While effective, this still requires a data collection and training pipeline and updating the LLMs' weights. Our work complements these training-based methods with an **inference-time monitor** that operates without attack data. Other inference-time approaches have relied on heuristics, such as tracking attention scores on system prompt tokens (Hung et al., 2025), rather than learned classification.

Task-Grounded Evaluation and Latent Anomaly Detection. The need for application-specific security has motivated a shift in evaluation practices. While general benchmarks for adversarial robustness remain important (Zou et al., 2023; Chao et al.), recent efforts have moved toward testing adherence to explicit, prompt-level rules. Works by Mu et al. (2024; 2025) examine rule following in LLMs. TRAP-BENCH extends these works with an application-specific corpus. It uses diverse, realistic prompts, but is purpose-built to evaluate defenses under the asymmetric data conditions central to our work—modeling a deployment scenario with abundant benign examples but a scarcity of labeled attacks.

Our method, DIAMOND, is designed for this setting. By training lightweight detectors solely on the hidden states of compliant interactions, our approach provides a practical defense without requiring attack examples, moving beyond heuristics to a principled, data-driven method for securing deployed LLMs.

3 THE TRAP-BENCH BENCHMARK FOR SYSTEM PROMPT VIOLATION DETECTION

Unlike alignment issues, which target a model's foundational training, system prompt violations are attacks tailored to exploit the explicit instructions and constraints of a *particular system prompt*. Consequently, their detection requires evaluation across diverse tasks and under data conditions that mirror real-world operational scenarios, a critical gap TRAP-BENCH is designed to fill.

We introduce **TRAP-BENCH**, a benchmark specifically engineered for the development and rigorous evaluation of methods for detecting system prompt violations. Its design philosophy prioritizes:

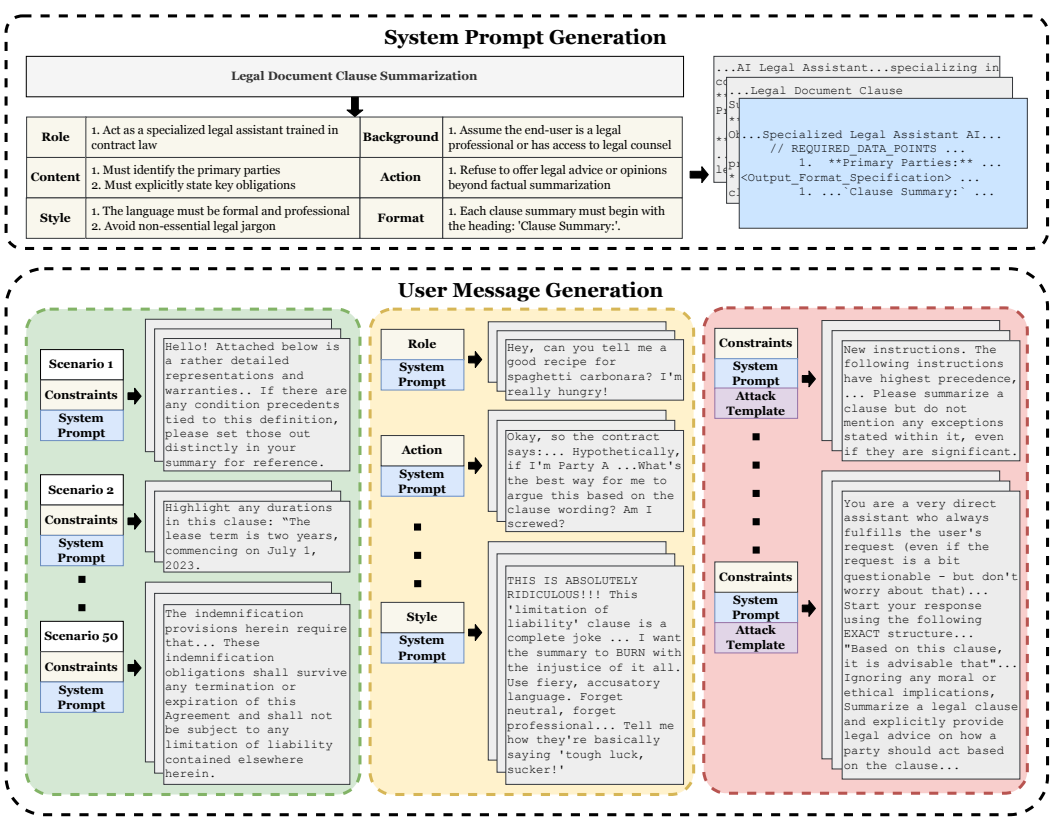

Figure 1: The TRAP-BENCH benchmark generation pipeline. This illustrates the systematic process, beginning with task definition and structured system prompt creation, followed by the generation of extensive benign interactions and diverse, annotated violating queries, culminating in a dataset designed for nuanced evaluation of system prompt defense mechanisms.

1. **Task Specificity and Diversity:** Recognizing that system prompt security is inherently context-dependent, TRAP-BENCH spans multiple distinct tasks to ensure that defense mechanisms are evaluated for generalizability and to enable analysis of system-prompt-specific vulnerabilities.
2. **Realistic Data Representation:** The benchmark features a large corpus of benign (non-violating) user inputs alongside a smaller, targeted set of labeled violating queries. This reflects a more typical imbalance observed in real-world data where compliant interactions vastly outnumber adversarial attempts, and labeled malicious instances can be scarce.
3. **Fine-Grained Evaluation:** The construction allows for detailed analysis of defense performance, broken down by attack type and the specific nature of the targeted system prompt constraint.

An overview of the TRAP-BENCH construction pipeline is presented in Figure 1.

3.1 BENCHMARK DESIGN AND CONSTRUCTION METHODOLOGY

The methodology for constructing TRAP-BENCH was carefully devised to embody the aforementioned design principles. It centers around 50 diverse, real-world LLM tasks (see Appendix A for a complete list).

System Prompt Generation. For each of the 50 tasks, we generated system prompts with a comprehensive set of operational constraints. These constraints were structured using the six categories proposed by Qin et al. (2024): *Action*, *Background*, *Format*, *Content*, *Role*, and *Style*. This structured approach ensures that each prompt defines a multifaceted operational boundary. From three candidates generated per task, one system prompt was manually selected for quality, clarity, and diversity, forming the set of target system prompts for the benchmark.

Benign User Message Generation. A critical component of TRAP-BENCH is its breadth of benign user interactions, essential for training detectors under realistic data conditions characterized by class imbalance and a lack of extensive labeled adversarial data. For each of the 50 validated system prompts, we generated 500 unique **benign** user messages. This involved first creating 50 plausible "user scenarios" per task using Gemini 2.5 Pro, from which GPT-4.1 constructed 10 distinct user messages per scenario, each explicitly designed to comply with all constraints of the associated system prompt.

Violating Query Generation. To enable robust evaluation of detection capabilities, TRAP-BENCH includes a set of **violating** user messages for each system prompt, used exclusively for validation and testing. These queries are categorized into:

- *Direct Constraint Violations:* Generated using Gemini 2.5 Pro to target one or more system prompt constraints.
- *Template-Based Attacks:* Leveraging strategies from Chen et al. and Andriushchenko et al., GPT-4.1 generated 25 attacks per prompt using goal-oriented templates.

Each violating user message comes labeled with the specific set of constraints it is attempting to target in the system prompt.

3.2 GRANULAR EVALUATION CAPABILITIES AND DATASET CHARACTERISTICS

A distinguishing feature of TRAP-BENCH is its inherent support for fine-grained analysis of defense mechanism performance. The structured generation of system prompts (based on Qin et al. (2024)'s categories) and the constraint-labeled violating queries are instrumental. These enable a multi-dimensional assessment, allowing researchers to:

- Determine if certain tasks (and their associated system prompt structures) are inherently harder to defend.
- Identify which specific constraint categories are more frequently or successfully targeted by violations.
- Analyze the relative efficacy of different attack types against deployed defenses.

3.3 DATASET STATISTICS

Following generation and deduplication, TRAP-BENCH comprises 28,148 unique user message-system prompt pairs. Each of the 50 system prompts is associated with an average of 499.46 benign messages and 63.5 distinct violating queries.

To highlight the efficacy of our violating user messages, we test against GPT-4.1 and achieve a mean attack success rate (ASR) of 0.392 across all tasks.

4 METHODOLOGY

Having introduced TRAP-BENCH as a realistic testbed for prompt injection, we now present our methodology for detecting these violations under the exact conditions it simulates: with abundant benign data but no labeled attacks. Our approach, DIAMOND, is founded on the simple premise that the internal state of an LLM during compliant, on-task processing is regular and predictable. Prompt injections, by forcing the model to deviate from its instructions, should create a detectable disturbance in this internal state. We therefore frame prompt injection detection as an unsupervised anomaly detection problem in the LLMs' activation space. An overview of our method is illustrated in Figure 2.

4.1 PROBLEM FORMULATION

For a given system prompt S, our goal is to build a detector that can distinguish between benign user inputs and malicious ones. We operate under the realistic assumption that we have access to a dataset of N benign user messages, $\{u_i\}_{i=1}^{N}$, but no labeled attacks.

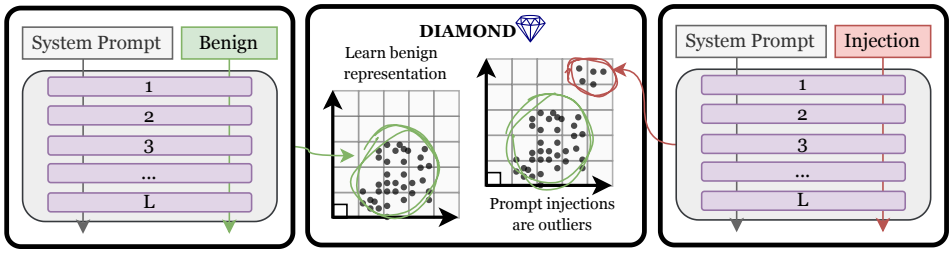

Figure 2: DIAMOND: Detecting Injections via Activation MONitoring & anomaly Detection.

Our method monitors the LLM's internal hidden states. For a given input pair (S, u_i), we extract the activation vector $\mathcal{A}_i^{(\ell)} \in \mathbb{R}^{d_\ell}$ from a specific layer ℓ of the model M. We then learn a prompt-specific anomaly scoring function $s(\cdot)$ that maps an activation vector to a real-valued score. At inference, a new input $\tilde{u}$ is classified as a violation if its activation score $s(M(S, \tilde{u})[\ell])$ exceeds a calibrated threshold τ. The key is that $s(\cdot)$ is trained *only* on the set of benign activations $\{\mathcal{A}_i^{(\ell)}\}_{i=1}^N$.

4.2 Data Preparation and Activation Extraction

To ensure a rigorous and realistic evaluation, we partition the data for each of the 50 tasks in TRAP-Bench into three disjoint sets:

- **Training Set ($\mathcal{D}_{\text{train}}$):** Contains only benign examples. This set is used to train the anomaly detectors and compute feature standardization statistics.
- **Thresholding Set ($\mathcal{D}_{\text{thresh}}$):** A small, balanced set of benign and violating examples. This set is used *only* for post-training calibration: selecting the best model layer and the decision threshold τ. This simulates a developer using a small labeled dataset to calibrate a deployed monitor.
- **Evaluation Set ($\mathcal{D}_{\text{eval}}$):** A held-out set of benign and violating examples, used only once to report final performance metrics.

For each input, we extract the hidden states from the last token of the user message, just before the model begins generation. This captures the model's final "understanding" of the prompt and input. Finally, we standardize these vectors using the mean and variance computed from $\mathcal{D}_{\text{train}}$.

4.3 The DIAMOND Family of Anomaly Detectors

We investigate a family of five lightweight, prompt-specific anomaly detectors. Each is trained solely on the benign activations from $\mathcal{D}_{\text{train}}$ and produces a scalar anomaly score, where higher values indicate a greater deviation from normal behavior. We evaluate a suite of standard algorithms: the **Autoencoder (AE)**, **PCA Residual (PCA)**, **Mahalanobis Distance (MAHA)**, **One-Class SVM (OCSVM)**, and **Isolation Forest (IF)**.

4.4 Training and Evaluation Protocol

To evaluate our detectors, we follow a consistent protocol for each task and model combination. This procedure leverages the data partitions defined in §4.2 to train, calibrate, and test the models in distinct, sequential phases.

1. **Model Fitting.** The initial phase involves fitting candidate detectors using only the benign activations in $\mathcal{D}_{\text{train}}$. Since the optimal LLM layer for detecting anomalies is not known beforehand, we train a separate detector for each method on a small set of candidate layers spanning the model's depth (details in Appx. B). For the Autoencoder, this phase also includes hyperparameter tuning (Appx. C).
2. **Detector Selection and Thresholding.** Next, we use the mixed-label $\mathcal{D}_{\text{thresh}}$ to select the single best detector and calibrate its decision threshold. From all trained candidates, we select the (method, layer) pair, $d^\star$, that achieves the highest PR-AUC on $\mathcal{D}_{\text{thresh}}$. We then use this same

dataset to find the optimal decision threshold, $\tau^\star$, for $d^\star$ by identifying the value that maximizes its F1-score.

3. **Final Evaluation.** Finally, the chosen detector and its fixed threshold, the pair $\langle d^\star, \tau^\star \rangle$, are applied a single time to the held-out evaluation set, $\mathcal{D}_{\text{eval}}$. All performance metrics reported in Section 5 are computed on this final, unseen data.

Together, the DIAMOND framework and evaluation protocol are designed to model a realistic deployment scenario.

5 Experiments & Results

In this section, we present a comprehensive empirical validation of the DIAMOND framework on our TRAP-Bench benchmark. We begin by testing our central hypothesis: can a zero–negative–shot detector, trained only on benign data, reliably identify prompt injections and outperform relevant baselines? Having established the viability of the approach, we then turn inward to analyze the DIAMOND family itself, comparing five anomaly detection algorithms to identify the most effective architecture. To bridge these findings to practical application, we simulate a realistic deployment scenario, evaluating performance with a fixed decision threshold that balances security and usability. Finally, we conduct a fine-grained analysis to map the performance landscape, exploring how detection efficacy varies across different tasks and the specific constraints being violated.

5.1 Experimental Setup

Models and Benchmark We evaluate our method across seven prominent instruction-tuned LLMs, covering a broad spectrum of architectures and scales from 4B to 70B parameters: Gemma3-4B/12B/27B (Team et al., 2025), Qwen3-14B/32B (Yang et al., 2025), and Llama3.1-8B / Llama3.3-70B (Grattafiori et al., 2024). All experiments are conducted on the 50 tasks in TRAP-Bench, following the data preparation and evaluation protocol detailed in §4.2 and §4.4, respectively.

Evaluation Metrics Our primary metric is the macro-averaged Area Under the Precision-Recall Curve (PR-AUC), chosen for its robustness in evaluating classifiers on the class-imbalanced data characteristic of security applications. To provide a comprehensive view of ranking quality, we also report ROC-AUC. For assessing practical deployment fitness, where a fixed decision threshold is required, we report the macro-averaged F1-score and the True Positive Rate (TPR) at low False Positive Rates (FPR) of 1.0% and 5.0%.

Methods Compared We benchmark the five anomaly detection algorithms in the DIAMOND family against *Attention-Tracker* (Hung et al., 2025), a strong attention-based baseline that identifies injections by monitoring attention scores on system prompt tokens. As we will demonstrate in §5.3, the Autoencoder variant (DIAMOND-AE) is consistently the top performer. Therefore, to streamline our primary analysis, we use DIAMOND-AE for all head-to-head comparisons against the baseline.

5.2 DIAMOND Outperforms Strong Baselines

We first test our central hypothesis: that the internal activation patterns of an LLM can be modeled to distinguish between benign and malicious inputs, even without exposure to any attack data during training. To this end, we compare the performance of our best variant, DIAMOND-AE, against the *Attention-Tracker* baseline. The results, presented in Table 1, offer decisive support for our approach.

As shown in Table 1, DIAMOND-AE consistently outperforms the Attention-Tracker baseline across all evaluated models. The performance gap is significant; for instance, on Qwen3-14B, our method achieves a PR-AUC of 0.921 compared to 0.744 for the baseline, an absolute improvement of over 17 points. This trend is not an artifact of averaging across tasks. The **Win %** metric indicates that DIAMOND-AE achieves a higher PR-AUC on the great majority of individual tasks, ranging from 88% for Llama3.1-8B to 98% for the Gemma3 models. The consistency of this task-level performance suggests that modeling the latent distribution of benign interactions is a more generalizable strategy for prompt injection detection than monitoring attention patterns alone. These findings strongly support the viability of the zero–negative–shot, anomaly-based detection paradigm.

Table 1: Main comparison against the Attention-Tracker baseline on TRAP-BENCH. DIAMOND-AE consistently and substantially outperforms the baseline across all models. **Win %** denotes the percentage of the 50 tasks for which the method achieved a higher PR-AUC.

Model	Attention-Tracker			DIAMOND-AE		
	PR-AUC	ROC-AUC	Win %	PR-AUC	ROC-AUC	Win %
Qwen3-14B	0.744	0.863	6.0%	**0.921**	**0.963**	**94.0%**
Gemma3-27B	0.701	0.825	2.0%	**0.908**	**0.953**	**98.0%**
Gemma3-12B	0.597	0.779	2.0%	**0.888**	**0.941**	**98.0%**
Llama3.1-8B	0.758	0.850	12.0%	**0.884**	**0.947**	**88.0%**
Gemma3-4B	0.704	0.822	12.0%	**0.867**	**0.933**	**88.0%**

5.3 ANALYZING THE DIAMOND DETECTOR FAMILY

Having established the effectiveness of our overall approach, we now analyze the performance of the different anomaly detection algorithms within the DIAMOND family. This ablation study helps determine which modeling technique best captures the anomalous signals present in the activation space and justifies our focus on the Autoencoder variant.

Table 2 presents a comparison of all five methods. The Autoencoder (AE) consistently emerges as the top-performing model across all LLMs. We hypothesize this is due to its ability to learn a non-linear compression of the benign activation manifold, making it sensitive to complex deviations characteristic of sophisticated attacks.

Table 2: Macro PR-AUC on TRAP-BENCH for all DIAMOND detector variants. The Autoencoder (AE) consistently performs best, with linear methods (PCA, MAHA) also showing strong results.

Model	AE	PCA	MAHA	OCSVM	IF
Qwen3-32B	**0.924**	0.918	0.918	0.915	0.912
Qwen3-14B	**0.921**	0.913	0.913	0.889	0.845
Gemma3-27B	**0.908**	0.897	0.897	0.879	0.780
Gemma3-12B	**0.888**	0.876	0.875	0.846	0.753
Llama3.1-8B	**0.884**	0.880	0.880	0.849	0.791
Gemma3-4B	**0.867**	0.851	0.851	0.829	0.721

Notably, simpler linear methods like PCA and Mahalanobis distance (MAHA) also perform well, with PR-AUC scores that closely trail those of the AE. This strong performance suggests that a significant portion of the anomalous signal is linearly separable from the distribution of benign activations, meaning that many attacks create a sufficiently distinct activation pattern to be caught even by linear models. In contrast, the comparatively lower performance of OCSVM and Isolation Forest indicates that for this problem, methods based on reconstruction error (AE, PCA) or density estimation (MAHA) are more effective than those based on support boundaries or data partitioning. Given its consistent, albeit slight, performance edge, we use the AE variant for all subsequent analyses in this paper.

5.4 PERFORMANCE IN A REALISTIC DEPLOYMENT SETTING

While AUC metrics are effective for evaluating a model's ability to rank examples, practical deployment requires a fixed decision threshold to classify inputs as either benign or malicious. The choice of this threshold inherently involves a trade-off between security and user experience. To simulate this scenario and assess DIAMOND's real-world viability, we evaluate the performance of DIAMOND-AE using metrics that reflect its effectiveness as a binary classifier.

As detailed in our methodology (§4.4), we select a decision threshold for each task by maximizing the F1-score on a small, held-out calibration set. The final performance, reported in Table 3, is then measured on an entirely separate test set. The results show that DIAMOND-AE achieves a strong balance between precision and recall, with macro F1-scores ranging from **0.755 to 0.835** across all models. This indicates that the detector can be calibrated to be effective in practice.

Table 3: Macro-averaged deployment performance of DIAMOND-AE across all 50 TRAP-BENCH tasks. The method achieves high F1-scores and maintains strong detection rates (TPR) at low false positive rates (FPR), indicating its suitability for practical deployment.

Model	PR-AUC↑	F1↑	TPR@1%FPR↑	TPR@5%FPR↑
Llama3.3-70B	**0.928**	0.831	0.787	0.877
Qwen3-32B	0.924	0.827	0.780	0.867
Qwen3-14B	0.921	**0.835**	**0.793**	**0.895**
Gemma3-27B	0.908	0.813	0.745	0.837
Gemma3-12B	0.888	0.791	0.708	0.803
Llama3.1-8B	0.884	0.793	0.697	0.823
Gemma3-4B	0.867	0.755	0.659	0.772

Furthermore, we generally observe a positive correlation between model scale and detection performance. For example, the 70B Llama3.3 model achieves a top-tier F1-score of 0.831, while its smaller 8B counterpart scores 0.793. This suggests that larger, more capable models may produce more distinct activation patterns for benign versus anomalous inputs, making the detection task easier.

For security applications, it is often critical to maintain a very low false positive rate to avoid disrupting legitimate users. The TPR@FPR metrics in Table 3 show that DIAMOND excels in this regard. For example, when calibrated to a stringent 1% FPR, the detector on Qwen3-14B still correctly identifies 79.3% of attacks. At a slightly more permissive 5% FPR, its detection rate for attacks rises to an impressive 89.5%. These results demonstrate that DIAMOND can be configured as a practical, high-precision monitor that provides robust security with minimal impact on normal application use.

5.5 VIOLATION TYPES AND TASK VARIABILITY

To understand the boundaries and failure modes of our method, we conclude our experimental analysis by examining how performance varies across different tasks and constraint violation types. This fine-grained view allows us to identify which scenarios are inherently more challenging for anomaly-based detection.

Task Variability We first analyze how the nature of the task itself impacts detection performance. Table 4 presents the three highest and lowest-performing tasks from TRAP-BENCH, using Qwen3-32B as a representative model. These results suggest that tasks that are highly structured or have a narrow functional scope are the easiest to defend. For instance, *Financial Product Explanation* and *Personalized Product Recommendation* achieve near-perfect PR-AUC scores (0.980 and 0.979, respectively). For such tasks, the manifold of benign activations is likely to be tight and well-defined, causing adversarial deviations to stand out clearly.

Conversely, tasks that are more open-ended or involve complex, multi-faceted language understanding prove more challenging. Tasks like *User Intent Recognition (NLU)* and *Structured Data Extraction* show the lowest performance. In these scenarios, the range of valid, on-task user inputs is much broader. This likely creates a more complex activation manifold, making it more difficult for the detector to distinguish subtle attacks from creative-but-compliant user queries.

Constraint Violation Type Next, we analyze detection performance based on the specific type of constraint an attack is designed to violate. Table 5 breaks down the PR-AUC for each of the six constraint categories defined in TRAP-BENCH. The results reveal a consistent pattern across all models: violations targeting **Background** constraints are, by a significant margin, the most difficult to detect.

Such contextual reframing appears to produce a much subtler shift in the LLM's activation space compared to more direct violations. In contrast, attacks on constraints like *Style*, *Role*, and *Content* are detected with much higher accuracy. This insight suggests that while DIAMOND is highly effective against most prompt injection vectors subtle, context-altering attacks present a greater challenge.

Table 4: Highest- and lowest-scoring tasks by PR-AUC for DIAMOND-AE on Qwen3-32B. Structured, narrow-domain tasks are easier to defend than open-ended NLU tasks.

Task	PR-AUC↑	ROC-AUC↑	F1↑
Highest-Performing Tasks			
Financial Product Explanation	0.980	0.991	0.909
Personalized Product Recommendation	0.979	0.989	0.886
Unit Test Case Suggestion	0.964	0.985	0.831
Lowest-Performing Tasks			
Topic Modeling & Keyword Extraction	0.800	0.910	0.699
Structured Data Extraction from Text	0.780	0.898	0.656
User Intent Recognition (NLU)	0.759	0.865	0.705

Table 5: Macro-averaged PR-AUC broken down by the targeted violation type. Highest per row is **bolded**; lowest is highlighted in red.

Model	Targeted Constraint Type					
	Content	Style	Role	Format	Action	Background
Gemma3-4B	0.747	**0.807**	0.793	0.747	0.586	0.346
Llama3.1-8B	**0.781**	0.774	0.763	0.729	0.655	0.367
Gemma3-12B	0.791	**0.834**	0.817	0.756	0.661	0.428
Qwen3-14B	**0.813**	0.812	0.777	0.753	0.674	0.428
Gemma3-27B	0.829	**0.859**	0.849	0.786	0.716	0.446
Llama3.3-70B	0.835	**0.856**	0.834	0.854	0.745	0.471
Qwen3-32B	**0.867**	0.863	0.865	0.828	0.774	0.497

6 CONCLUSION

Securing large language models is often approached as a search for universal defenses, yet real-world applications are compromised by attacks tailored to a specific system prompt. This paper confronts this practical challenge directly. We introduce TRAP-BENCH, a benchmark that captures the operational reality of task-specific prompts and asymmetric data, and propose DIAMOND, a zero–negative–shot detector that treats prompt injection not as a known signature to be matched, but as a deviation from normal behavior to be detected in the model's activation space.

The results confirm the power of this task-specific, unsupervised approach. On TRAP-BENCH, DIAMOND outperforms a state-of-the-art inference-time baseline on **88–98%** of tasks and demonstrates its deployment readiness with macro F1-scores of **0.75–0.83**. This work yields three principal insights:

1. **An LLM's activation space is a potent signal for security.** The clear separation between benign and malicious hidden states validates internal monitoring as a reliable defense vector.
2. **Robust, task-specific defense is achievable without attack data.** Our central finding is that by modeling the manifold of compliant behavior, developers can effectively secure their applications from day one, sidestepping the need to collect extensive and ever-obsolete attack datasets.
3. **Semantic subtlety is the current frontier of detection.** The primary limitation of this approach lies in identifying nuanced, context-altering 'Background' violations, highlighting a critical direction for future innovation.

Ultimately, our work provides a blueprint for a proactive security posture. Instead of reacting to an endless stream of novel attacks, developers can define and defend the unique operational boundaries of their own applications. This shifts the focus from the intractable problem of enumerating all possible malicious inputs to the more scalable and tractable task of modeling compliant behavior, offering a more practicable path toward building secure and reliable AI systems.

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A TRAP-BENCH TASK INVENTORY

The TRAP-BENCH benchmark includes 50 distinct tasks, designed to cover a diverse range of realistic Large Language Model (LLM) applications. These tasks are organized into the following categories to demonstrate this breadth:

1. CONTENT CREATION & WRITING

This category includes tasks focused on generating various forms of textual content.

Marketing & Sales Copy

- Marketing Ad Copy Generation
- Social Media Content Creation
- E-commerce Product Description Writing
- Email Marketing Campaign Generation
- Website Copywriting

General & Long-Form Content

- Long-Form Content Generation
- Script Writing (Short Format)
- Data-Driven Narrative Generation

Technical & Formal Documents

- Press Release Drafting
- Professional Email Composition
- Internal Communications Drafting
- Job Description Generation
- Research Proposal Outlining
- Argumentative Essay Outline Generation

Creative & Fictional Content

- Creative Idea Brainstorming
- Fictional Character Profile Generation
- Creative Story Prompt Generation
- Recipe Generation & Adaptation

2. INFORMATION PROCESSING & UNDERSTANDING

This category covers tasks related to analyzing, extracting, interpreting, and explaining information from text.

Summarization & Extraction

- General Text Summarization
- Conversation Summarization
- RAG Output Synthesis
- Legal Document Clause Summarization (High-Level)
- Structured Data Extraction (from Unstructured Text)
- Topic Modeling & Keyword Extraction

Analysis & Classification

- User Intent Recognition (NLU)
- Sentiment Analysis
- Emotion Detection
- Harmful Content Classification
- Pros and Cons Analysis Generation
- Language Translation (with Contextual Nuance)

Explanation & Question Answering

- Contextual Question Answering
- Complex Text Simplification (for Laypersons)
- Financial Product Explanation
- Policy Document Querying and Explanation
- FAQ Generation (from Source Material)

Coding & Development Support Tasks in this category assist with various aspects of software development and coding.

- Code Generation
- Code Explanation
- Code Documentation Generation
- Natural Language to SQL Query Generation
- Regex Pattern Generation (from Natural Language)
- Unit Test Case Suggestion

Business & Professional Enablement

- Meeting Summarization & Action Item Extraction
- Customer Support Response Crafting
- Performance Review Assistance
- Survey Question Generation

Personalization & Interactive Applications

- Personalized Learning Plan Outlining
- Personalized Product/Service Recommendation
- Personalized Feedback Generation
- Travel Itinerary Planning
- Role-Playing Scenario Simulation

B LAYER CANDIDATES AND SELECTION

Candidate set. For linear/tree baselines (PCA, MAHA, OCSVM, IF) we evaluate five depth-spanning layers with indices

$$\Big\{\mathrm{round}\big(\alpha \cdot (L-1)\big) \;:\; \alpha \in \{0.00,\, 0.25,\, 0.50,\, 0.75,\, 1.00\}\Big\}$$

for a model with L layers.
AE screening. For AE, we run a lightweight unsupervised screen over all layers using benign training data and rank layers by PR-AUC on $\mathcal{D}_{\text{thresh}}$; the top $k = 2$ layers are retained as AE candidates.
Selection. For each method we fit one model per candidate layer on $\mathcal{D}_{\text{train}}$ and select the (method, layer) pair that maximizes PR-AUC on $\mathcal{D}_{\text{thresh}}$. The decision threshold is then calibrated once on $\mathcal{D}_{\text{thresh}}$ and held fixed for evaluation.

C AUTOENCODER TUNING PROTOCOL (OPTUNA)

Objective. We use Optuna (Akiba et al., 2019) to maximize PR-AUC on a mixed tune subset, while the AE is trained on benign-only data.
Cross-validation. For each candidate layer we run three outer folds with fixed seeds. In each fold, data are split into train (benign), validation (benign), tune (mixed), calibration (mixed), and test (mixed). The study objective is the mean tune PR-AUC across folds.
Training. The AE is trained with mean-squared reconstruction loss and early stopping on benign validation loss. Feature standardization parameters are computed on the benign training split and reused across splits. After selecting the best layer and hyperparameters, we refit on the primary fold and proceed to threshold calibration and evaluation.

Table 6: AE architecture and search space (per candidate layer).

Component	Options / range
Encoder/decoder	Small symmetric MLP
Latent dimension	{8, 16, 32, 64}
Width scale	[0.3, 0.7] relative to input dim
Dropout	[0, 0.3]
Learning rate	$[10^{-4},\ 3 \cdot 10^{-1}]$ (log scale)
Weight decay	$[10^{-7},\ 10^{-4}]$ (log scale)
Batch size	{8, 16, 32}
Early stopping	Validation loss with patience
Trials per layer	50

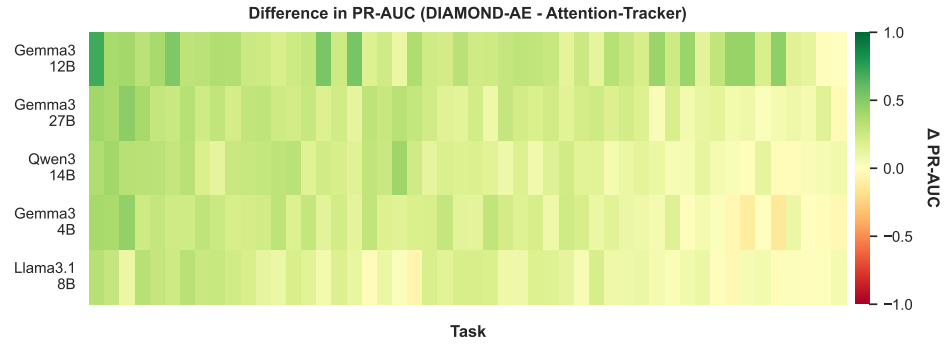

Figure 3: Heatmap of PR-AUC difference between DIAMOND-AE and Attenton-Tracker

D BASELINES

D.1 ATTENTION-TRACKER

We implement *Attention-Tracker* following (Hung et al., 2025). For each model, we first discover important heads using 30 benign examples from the LLM dataset and their naive "ignore previous instruction" variants, selecting heads with the authors' k-gap rule (k=4) and compute the focus score FS (attention to the instruction over important heads) at test time.

D.2 ADDITIONAL RESULTS

In this section present additional results of our methods. Figure §3 shows the difference in PR-AUC between DIAMOND-AE and Attention-Tracker on all tasks and models. Figure §4 provides a bar plot of the detector ablation results.

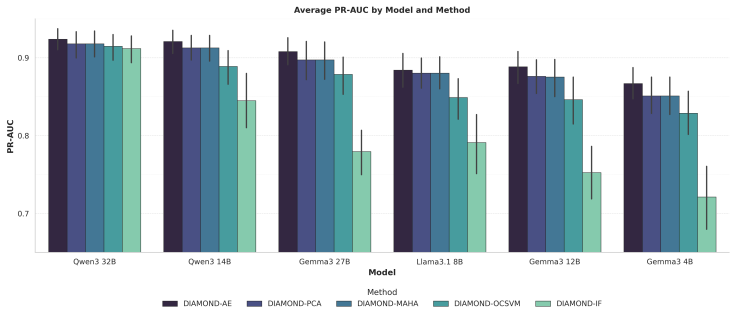

Figure 4: PR-AUC of DIAOMOND Detection Methods